

- Navigate to:
- Home
 - Customer Segments
 - Churn Prediction
 - Business Insights

Dataset: Online Retail II
Period: Dec 2009 – Dec 2011
Customers: 5,878
Transactions: 779,425

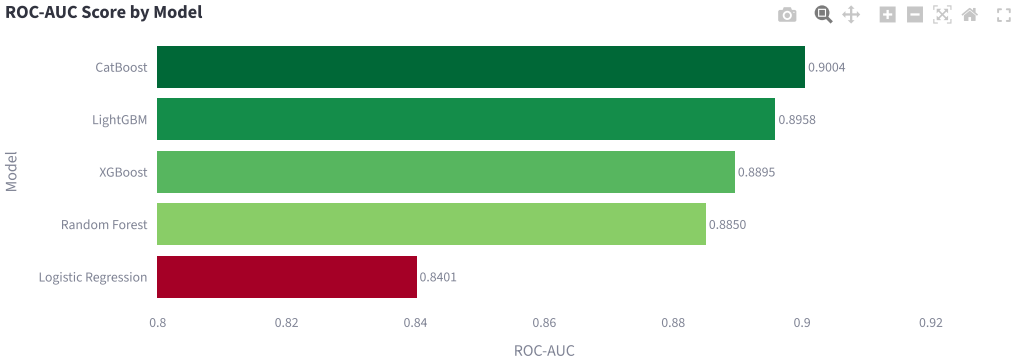
Model: CatBoost Classifier
ROC-AUC: 0.90
F1 Score: 0.80

Business Insights

Revenue trends, customer behavior, and model performance

- Revenue Analysis Geographic Analysis Model Performance

Model Comparison — All 5 Models



Detailed Metrics Table

	Model	ROC-AUC	F1	Accuracy	Precision	Recall
0	CatBoost	0.9004	0.8000	0.8010	0.8182	0.7826
1	LightGBM	0.8958	0.7898	0.7891	0.8007	0.7793
2	XGBoost	0.8895	0.7790	0.7781	0.7890	0.7692
3	Random Forest	0.8850	0.7833	0.7840	0.7997	0.7676
4	Logistic Regression	0.8401	0.7639	0.7466	0.7259	0.8060

Best Model: CatBoost

- ROC-AUC: 0.9004
- F1 Score: 0.8000
- Precision: 0.8182
- Recall: 0.7826
- PR-AUC: 0.9090

Why CatBoost Won

- Handles categorical features natively
- Built-in regularization reduces overfitting
- Best precision — minimizes false alarms
- Consistent across all cross-validation folds